Chapter 51 — Runtime Permissions Management

> **Part VI — Lifecycle, Background & Platform** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 44 (lifecycle), Ch 6 (UI state), Ch 49/50 (notifications), Ch 67 (location), Ch 25 (storage). Pairs with Ch 52 (deep links to settings), Ch 28 (privacy).
>
> **Reference surface in StreamCart:** requesting and managing dangerous permissions — notifications (Android 13+), camera (scan a product/QR), location (nearby stores/ delivery), media/photos (upload a review image), microphone (voice search) — with correct rationale, graceful denial handling, and no feature breakage when denied.

51.1 Problem Definition and Real-World Context

Android gates access to sensitive capabilities (camera, location, contacts, mic, notifications, media) behind **runtime permissions** the user grants or denies in a system dialog. This is a **consent and UX contract**, and it's easy to get wrong in ways that either break features or get the app rejected/uninstalled:

- **Asking at the wrong time.** Requesting camera/location at app launch (before the user understands why) → reflexive denial, and now the feature is dead. Permissions must be requested **in context**, when the user triggers the feature.
- **No rationale.** The system dialog is terse; without an app-provided **rationale** (especially after a first denial), users don't understand the value and deny.
- **Mishandling the denial states.** There are several: not-yet-asked, granted, denied (can ask again), and **permanently denied** ("don't ask again" / denied twice on modern Android — the system dialog won't show again). Treating "denied" as one state, or trying to re-request when permanently denied (silently does nothing), breaks the flow.
- **Not degrading gracefully.** A denied permission should **disable just that feature** with a clear path to settings — not crash, not block the whole app, not nag.
- **New permission models. Notifications** became a runtime permission (Android 13+); **location** has foreground/background split + "approximate vs precise" + "only this time"; **media** moved to granular `READ_MEDIA_IMAGES/VIDEO/AUDIO` and the **Photo Picker** / "Selected photos" partial access (Android 13/14). Old patterns (request all storage) are wrong now.
- **Permissions can be revoked** anytime (settings, or auto-reset for unused apps, Android 11+) — so you must **check at use**, never cache "granted" forever.

The architecture: **request in context** (on feature trigger, not launch); **show rationale** (before, or after a denial, via `shouldShowRequestPermissionRationale`); use the **Activity Result API** (`registerForActivityResult`) — not the deprecated `onRequestPermissionsResult`; **model the full state machine** (not-asked / granted / denied-can-ask / permanently-denied) and route each correctly (request / proceed / rationale+request / deep-link to settings, Ch 52); **check permission at use** (it can be revoked); **degrade gracefully** (disable just the feature); and use the **modern privacy-friendly APIs** (Photo Picker, approximate location, "only this time"). This chapter is about **getting consent respectfully and handling every outcome without breaking the app**.

Three practical use cases

1. **Notifications (Android 13+).** Request `POST_NOTIFICATIONS` after onboarding / when the user opts into alerts — with rationale — so pushes (Ch 49/50) can display; if denied, the inbox still works (Ch 50).
2. **Camera for QR/product scan (StreamCart).** Request `CAMERA` when the user taps "Scan"; rationale if previously denied; on permanent denial, show a settings deep-link; feature disabled cleanly otherwise.
3. **Photos for a review image.** Use the **Photo Picker** (no permission needed) by default; only request `READ_MEDIA_IMAGES` if full-gallery access is truly required, handling "Selected photos" partial grants.

Why this is architectural

Permission handling is a **cross-cutting consent state machine** touching lifecycle (Ch 44), UI state (Ch 6), many features (camera/location/media/notifications), privacy (Ch 28), and deep links to settings (Ch 52). The decisions — when to ask, how to explain, how to handle each outcome, how to degrade — determine grant rates, feature reliability, and user trust.

51.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Request permissions in context (on feature trigger).
F2	Show rationale before/after denial to explain value.
F3	Model and route all states (not-asked/granted/denied/permanently-denied).
F4	Check permission at use (revocation/auto-reset).
F5	Degrade gracefully (disable just the feature; settings path).
F6	Use modern privacy APIs (Photo Picker, approximate location, partial media).

Non-functional requirements

Attribute	Policy
UX / grant rate	In-context + rationale; never nag; clear value.
Correctness	Check at use; handle permanent denial (no silent re-request).
Resilience	No crash/whole-app block on denial.
Privacy	Least privilege; prefer no-permission APIs (Photo Picker); minimal scope.
Maintainability	Central permission helper + state machine, not ad hoc.

Scope

- **In scope:** runtime permission model, dangerous vs normal permissions, Activity Result API, rationale (`shouldShowRequestPermissionRationale`), state machine (not-asked/granted/denied/permanently-denied), check-at-use, graceful degradation, settings deep-link (Ch 52), notifications (13+), location (fg/bg, approximate/precise, one-time), media (granular + Photo Picker + partial), auto-reset, central helper.
- **Out of scope:** specific feature internals (camera/location — Ch 67), privacy policy (Ch 28), deep-link mechanics (Ch 52), notification rendering (Ch 50).

Assumptions & constraints

- `minSdk 24`, `targetSdk` current (13+ notifications, 14 partial media). AndroidX Activity Result API, Compose (`rememberLauncherForActivityResult` / Accompanist-style patterns). Permissions can be revoked anytime; permanent denial has no programmatic re-prompt (only settings).

51.3 System Design Fundamentals

- **Normal vs dangerous permissions. Normal** (e.g., `INTERNET`) are granted at install, declared in the manifest. **Dangerous** (camera, location, contacts, mic, notifications, media) require a **runtime request** and user consent. This chapter is about dangerous permissions.
- **The permission state machine (model all of it):**
  1. **Not requested** → request (optionally with pre-rationale).
  2. **Granted** → proceed.
  3. **Denied (can ask again)** → `shouldShowRequestPermissionRationale()` returns **true** → show **rationale**, then re-request.
  4. **Permanently denied** ("don't ask again" / denied the required number of times) → `shouldShowRequestPermissionRationale()` returns **false** *and* not granted → the system dialog **won't show**; you must **deep-link to app settings** (Ch 52) and explain. Re-requesting does nothing (returns denied immediately). Collapsing these into "granted/denied" is the #1 bug — you must distinguish "denied-can-ask" from "permanently-denied" to decide between rationale+request and a settings deep-link.
- **Activity Result API (the correct mechanism).** Use `registerForActivityResult(RequestPermission()/RequestMultiplePermissions())` (Views) or `rememberLauncherForActivityResult` (Compose) — **not** the deprecated `requestPermissions` / `onRequestPermissionsResult`. It's lifecycle-safe (Ch 44) and returns the grant result in a callback.
- **Request in context, with rationale.** Ask when the user **triggers the feature** (taps "Scan", "Find nearby", "Enable alerts"), not at launch. If a previous denial occurred (`shouldShowRequestPermissionRationale` true), show a **rationale UI** first explaining the value, then request. This maximizes grant rate and respects the user.
- **Check at use, not once.** Permissions can be **revoked** in settings or **auto-reset** for unused apps (Android 11+). Always `ContextCompat. checkSelfPermission()` **at the point of use**; never assume a past grant persists.
- **Graceful degradation.** A denied permission disables **only that feature**, with a clear message and (if permanently denied) a settings path — the rest of the app works. Never crash, never block the whole app,

never nag repeatedly (one in-context ask + rationale; after that, a passive settings link).

- **Modern per-permission specifics:**
  - **Notifications (`POST_NOTIFICATIONS`, Android 13+):** runtime permission; request when the user opts into alerts / after onboarding. Denied → system notifications suppressed, but the **inbox still works** (Ch 50). Don't request at first launch before value is shown.
  - **Location:** foreground (`ACCESS_FINE_LOCATION` / `ACCESS_COARSE_LOCATION`) vs **background** (`ACCESS_BACKGROUND_LOCATION`, must be requested **separately/after** foreground, and sparingly, Ch 67); **approximate vs precise** (Android 12+ — request coarse if precise isn't needed); **"only this time"** grants (ephemeral). Request the least you need.
  - **Media (Android 13+):** granular `READ_MEDIA_IMAGES` / `VIDEO` / `AUDIO` replaced `READ_EXTERNAL_STORAGE`; **Android 14** adds **partial access** ("Selected photos" → `READ_MEDIA_VISUAL_USER_SELECTED`). **Prefer the Photo Picker** (`PickVisualMedia`) which needs **no permission** at all — use it unless you truly need the whole library.
  - **Camera/Mic:** request on feature use; nothing special beyond the state machine.
- **Manifest + policy:** declare each permission; Play requires a **declared, justified** use for sensitive ones (location/camera/mic background) and may reject over-broad requests. Least privilege is both UX and policy.

Core insight: **request dangerous permissions in context (on feature trigger) with a clear rationale, using the Activity Result API; model the full state machine — not-asked / granted / denied-can-ask (rationale + re-request) / permanently-denied (settings deep-link, no silent re-request) — distinguishing the last two via `shouldShowRequestPermissionRationale`; check the permission at the point of use (it can be revoked/auto-reset); degrade gracefully (disable just the feature, never crash/nag); and prefer the least-privilege, no-permission modern APIs (Photo Picker, approximate location, one-time grants).**

51.4 High-Level Architecture


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Components & ownership

- **Feature trigger** initiates an in-context check.
- **State machine** routes to proceed / request / rationale+request / settings.
- **Activity Result launcher** performs the request lifecycle-safely.
- **Graceful degradation** disables just the feature on denial.
- **No-permission APIs** (Photo Picker) bypass the whole flow when possible.

Direction: trigger → check-at-use → state-routed request/rationale/settings → proceed or degrade.

51.5 Detailed Component Design


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Component	Responsibility
`PermissionState`	the 4-state model (the crux)
`PermissionController`	compute state, launch request, open settings (Ch 52)
`FeatureGate`	gate a feature on grant; degrade on denial
Rationale UI	explain value (before/after denial)

51.6 End-to-End Data Flow

In-context request with rationale and permanent-denial handling



Media: prefer Photo Picker (no permission)



51.7 Complete Kotlin Implementation

Package: `com.streamcart.permissions`.

51.7.1 The state model + controller

```
sealed interface PermissionState {
    data object Granted : PermissionState
    data object NotRequested : PermissionState
    data object DeniedCanAsk : PermissionState       // shouldShowRationale == true
    data object PermanentlyDenied : PermissionState  // asked, not granted, rationale == false
}

class PermissionController(private val activity: ComponentActivity) {
    fun state(permission: String, alreadyAsked: Boolean): PermissionState = when {
        ContextCompat.checkSelfPermission(activity, permission) ==
PackageManager.PERMISSION_GRANTED ->
            PermissionState.Granted
        activity.shouldShowRequestPermissionRationale(permission) ->
PermissionState.DeniedCanAsk
        alreadyAsked -> PermissionState.PermanentlyDenied   // asked before + no rationale + not
granted
        else -> PermissionState.NotRequested
    }
    fun openAppSettings() = activity.startActivity(                 // Ch 52
        Intent(Settings.ACTION_APPLICATION_DETAILS_SETTINGS,
Uri.parse("package:${activity.packageName}")))
}
// Persist "alreadyAsked" per permission (DataStore, Ch 4) since the system doesn't expose it
directly.
```

51.7.2 Compose: in-context request, rationale, degrade

```
@Composable
fun ScanButton(viewModel: ScanViewModel = hiltViewModel()) {
    val context = LocalContext.current
    var showRationale by remember { mutableStateOf(false) }
    var showSettings by remember { mutableStateOf(false) }

    val launcher =
rememberLauncherForActivityResult(ActivityResultContracts.RequestPermission()) { granted ->
        viewModel.onCameraResult(granted)   // granted -> open scanner; denied -> degrade
    }

    fun onScanClicked() {
        when {
            ContextCompat.checkSelfPermission(context, Manifest.permission.CAMERA)   // CHECK AT
USE
                == PackageManager.PERMISSION_GRANTED -> viewModel.openScanner()
            (context as
Activity).shouldShowRequestPermissionRationale(Manifest.permission.CAMERA) ->
                showRationale = true                                  // DeniedCanAsk ->
rationale first
            viewModel.cameraAlreadyAsked() -> showSettings = true    // PermanentlyDenied ->
settings
            else -> { viewModel.markCameraAsked(); launcher.launch(Manifest.permission.CAMERA) }
// first ask
        }
    }

    Button(onClick = ::onScanClicked) { Text("Scan") }

    if (showRationale) RationaleDialog(
        text = "Scanning uses your camera to read product and QR codes.",
        onContinue = { showRationale = false; viewModel.markCameraAsked();
launcher.launch(Manifest.permission.CAMERA) },
        onDismiss = { showRationale = false })
    if (showSettings) SettingsDialog(
        text = "Camera is turned off. Enable it in Settings to scan.",
        onOpenSettings = { showSettings = false; PermissionController(context as
ComponentActivity).openAppSettings() },
        onDismiss = { showSettings = false })
}
```

51.7.3 Notifications permission (Android 13+)

```
@Composable
fun EnableAlerts(onResult: (Boolean) -> Unit) {
    val launcher =
rememberLauncherForActivityResult(ActivityResultContracts.RequestPermission(), onResult)
    LaunchedEffect(Unit) {
        if (Build.VERSION.SDK_INT >= 33) launcher.launch(Manifest.permission.POST_NOTIFICATIONS)
        else onResult(true)   // pre-13: granted by manifest
    }
    // Request after onboarding / when user opts into alerts — NOT at first launch.
    // Denied -> inbox still works (Ch 50); don't block.
}
```

51.7.4 Media: Photo Picker (no permission) preferred

```
@Composable
fun AddReviewPhoto(onPicked: (Uri?) -> Unit) {
    // Photo Picker needs NO permission and is privacy-friendly (Android 13+, backported via
Play services).
    val picker = rememberLauncherForActivityResult(
        ActivityResultContracts.PickVisualMedia()) { uri -> onPicked(uri) }
    Button(onClick = {
picker.launch(PickVisualMediaRequest(ActivityResultContracts.PickVisualMedia.ImageOnly)) }) {
        Text("Add photo")
    }
    // Only fall back to READ_MEDIA_IMAGES (handling READ_MEDIA_VISUAL_USER_SELECTED partial) if
you
    // genuinely need full-library access (e.g., a gallery browser).
}
```

51.7.5 Location: least privilege (approximate, foreground-first)

```
// Request COARSE if precise isn't needed; request BACKGROUND only separately/after foreground
(Ch 67).
val locationLauncher = rememberLauncherForActivityResult(
    ActivityResultContracts.RequestMultiplePermissions()) { grants ->
    val coarse = grants[Manifest.permission.ACCESS_COARSE_LOCATION] == true
    viewModel.onLocationResult(precise = grants[Manifest.permission.ACCESS_FINE_LOCATION] ==
true, coarse = coarse)
}
// locationLauncher.launch(arrayOf(ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION))  // user may
pick "Approximate"
```

51.7.6 Module structure

```
:core:permissions  PermissionState, PermissionController, FeatureGate, rationale/settings UI
# "alreadyAsked" flags in DataStore (Ch 4). Settings deep-link -> Ch 52. Feature code per
feature module.
```

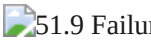
51.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Cached "granted" then revoked	crash/ `SecurityException` at use	check at use (not cached)
Re-request when permanently denied	nothing happens (silent)	detect state → settings deep-link
Launcher registered after STARTED	crash	register at composition/ `onCreate` (Activity Result API handles)
Config change mid-dialog	lost result	Activity Result API is lifecycle-safe (Ch 44)
Returning from settings	stale UI state	re-check permission on `ON_RESUME` (Ch 44)
Partial media grant treated as full	missing photos	handle `READ_MEDIA_VISUAL_USER_SELECTED` / use Photo Picker

- **Check at use + re-check on resume:** because permissions can be revoked (settings/ auto-reset) and the user may grant via settings then return, verify with `checkSelfPermission` at the point of use and re-evaluate on `ON_RESUME` (Ch 44) so the UI reflects a grant made in settings.
- **Activity Result API is lifecycle-safe:** it survives config changes and delivers the result correctly, unlike the deprecated `onRequestPermissionsResult` — register it at composition/ `onCreate`, never conditionally.

51.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Low grant rate	analytics	request in-context + rationale; don't ask at launch	higher grants
Permanent denial, feature dead	user stuck	detect state → settings deep-link + explain	recoverable
Crash on revoked permission	`SecurityException`	check at use; try/catch	no crash
Whole app blocked on denial	user report	degrade just the feature	app usable
Nagging repeated prompts	opt-out/uninstall	one ask + rationale, then passive settings link	retained
Over-broad request rejected by Play	review	least privilege; Photo Picker; justify	approved
Partial media missing files	user report	handle partial / use Photo Picker	correct


51.9 Failure Scenarios and Recovery

> **The two mistakes that define bad permission UX:** asking out of context (low grants) and mishandling permanent denial (feature permanently dead with no path out). Fix both with in-context rationale and a settings deep-link, and always degrade just the one feature.

51.10 Scaling the Implementation

- **Level 1.** Activity Result API; in-context request; check at use.
- **Level 2.** Full 4-state machine with rationale + settings deep-link; graceful degradation per feature.
- **Level 3 — production.** Per-permission "alreadyAsked" persistence; re-check on resume; notifications (13+), least-privilege location, Photo Picker for media.
- **Level 4 — large.** Central `PermissionController` / `FeatureGate` reused across features; grant-rate analytics + rationale A/B; auto-reset handling; consistent rationale/settings UI.
- **Level 5 — enterprise.** Permission governance (which features may request what, Play justification), privacy review (Ch 28), grant-rate dashboards, consent center, per-market/regulatory variations.

Avoid overengineering: a small central helper + the 4-state machine covers nearly everything. Don't wrap every permission in bespoke logic or request permissions you can avoid (use the Photo Picker and approximate location).

51.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
When to ask	In context (feature trigger)	at launch	In context (far higher grant rate)
Mechanism	Activity Result API	deprecated `onRequestPermissionsResult`	Activity Result API (lifecycle-safe)
State model	4-state machine	granted/denied	4-state (must distinguish permanent denial)
Media access	Photo Picker (no permission)	`READ_MEDIA_IMAGES`	Photo Picker unless whole library needed
Location precision	Approximate (coarse)	precise (fine)	Approximate unless precise truly needed
Permission caching	Check at use	cache granted	Check at use (revocation/auto-reset)
Denial handling	Degrade just the feature	block/crash/nag	Degrade (app remains usable)

In-context vs launch-time requests: asking at launch, before the user sees the feature's value, yields reflexive denials and often permanent denial — killing the feature. Requesting when the user actively triggers the feature

(with a rationale when previously denied) dramatically improves grant rates and respects the user. Always ask in context.

Photo Picker vs media permissions: the Photo Picker (`PickVisualMedia`) lets the user pick specific images **without any permission** and is the privacy-friendly default (backported via Play services). Requesting `READ_MEDIA_IMAGES` gives full-library access but triggers a permission dialog, partial-access complexity (Android 14 "Selected photos"), and Play scrutiny. Use the Photo Picker unless you genuinely need to browse the whole library.

Approximate vs precise location: many features (nearby stores, city-level content) work fine with **coarse** location, which users grant more readily and which is more privacy-respecting. Request **precise** only when truly needed (turn-by-turn), and **background** location only separately and sparingly (Ch 67) — over-requesting hurts grant rates and Play approval.

51.12 Security & Privacy Considerations

- **Least privilege:** request the **minimum** permission that enables the feature (coarse over fine, Photo Picker over full media, no background location unless essential). This is both privacy and Play policy (Ch 28).
- **Justify sensitive permissions:** Play requires declared, justified use for location/camera/mic (especially background); over-broad or unjustified requests are rejected. Provide an in-app rationale and a privacy policy.
- **Don't abuse rationale/prompts** to coerce; respect "no" (degrade, don't nag).
- **Handle "only this time" and partial grants** honestly — don't try to escalate an ephemeral/partial grant into persistent/full access.
- **Auto-reset for unused apps (Android 11+)** revokes permissions of unused apps; re-request in context when the feature is used again (don't assume persistence).
- **No sensitive data from a denied-then-revoked state:** check at use to avoid `SecurityException` and avoid leaking that a capability is unavailable in a way that harms UX.
- Client vs backend: permissions are a client consent concern; the backend should never assume the client has a capability (e.g., don't require location if denied — degrade server-side features too).

51.13 Performance and Resource Optimization

- Permission checks are **cheap** — do them at use; the cost is the user flow, not CPU.
- **Avoid unnecessary permissions entirely** (Photo Picker, approximate location, system share) — the best optimization is not needing the permission.
- **Don't block startup** on permission requests (Ch 45) — request in context later.
- **Re-check on resume** cheaply (Ch 44) rather than polling.
- **Batch multi-permission requests** (`RequestMultiplePermissions`) when a feature needs several at once (e.g., fine+coarse location) to avoid sequential dialogs.
- **Profiling:** grant/denial/permanent-denial rates per permission; feature-usage drop after denial; settings-deep-link conversion.

51.14 Testing Strategy

Permissions test with Espresso + `UiAutomator` / `GrantPermissionRule` (instrumentation) and unit tests for the state machine.

```
class PermissionStateTest {
    @Test fun `granted when checkSelfPermission granted`() { /* ... */ }
    @Test fun `DeniedCanAsk when shouldShowRationale true`() { /* ... */ }
    @Test fun `PermanentlyDenied when asked, not granted, no rationale`() { /* ... */ }
    @Test fun `NotRequested when never asked`() { /* ... */ }
}

@RunWith(AndroidJUnit4::class)
class ScanPermissionFlowTest {
    @get:Rule val grant = GrantPermissionRule.grant(Manifest.permission.CAMERA)
    @Test fun `scanner opens when camera granted`() { /* launch, tap Scan, assert scanner */ }
    // Denial/permanent-denial flows: use UiAutomator to deny in the system dialog and
    // assert rationale / settings dialog appears; verify feature degrades (no crash).
}
```

What to test: the 4-state computation; in-context flow (granted → proceed; denied → rationale; permanent → settings); check-at-use (revoke → re-enter flow); degrade (denied → feature disabled, no crash); notifications 13+; Photo Picker needs no permission; re-check on resume after settings. UiAutomator for real system-dialog interaction.

51.15 Production Observability

- **Metrics:** per-permission **grant / denied / permanently-denied** rates; in-context request rate; rationale-shown → grant conversion; settings-deep-link conversion; feature-usage after denial; auto-reset re-request rate; notification-permission grant (Ch 49/50).
- **Logs:** `permission{name, state, result, in_context}` (no PII, Ch 28).
- **Debugging:** low grants → asking out of context / weak rationale; feature "broken" → permanent denial not routed to settings; crashes → not checking at use; Play rejection → over-broad/unjustified.
- **Alerting:** grant-rate drop after a release (changed flow) or OS update (new model); permanent-denial spike.
- **Dashboards:** grant funnel per permission; feature retention post-denial.

51.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Normal vs dangerous?* Install-time (manifest) vs runtime consent.
2. *Mechanism?* Activity Result API (`registerForActivityResult` / `rememberLauncherForActivityResult`).
3. *Why not* `onRequestPermissionsResult` *?* Deprecated, not lifecycle-safe.
4. *When to request?* In context (feature trigger), not launch.
5. *Why rationale?* Explain value → higher grant rate.

State machine

6. *The states?* Not-asked / granted / denied-can-ask / permanently-denied.
7. *Distinguish denied vs permanent?* `shouldShowRequestPermissionRationale` (true = can ask; false + asked + not granted = permanent).
8. *Permanent denial handling?* Settings deep-link (Ch 52); re-request does nothing.
9. *Why persist "alreadyAsked"?* System doesn't expose not-asked vs permanent directly.

Check at use

10. *Cache granted?* No — can be revoked / auto-reset.
11. *Auto-reset?* Android 11+ revokes unused apps' permissions.
12. *Re-check when?* At use + on resume (after settings).

Degradation

13. *On denial?* Disable just that feature; never crash/block/nag.
14. *Settings path?* Deep-link to app settings with explanation.

Modern APIs

15. *Notifications?* `POST_NOTIFICATIONS` runtime (13+); request on opt-in; inbox still works if denied (Ch 50).
16. *Media?* Prefer Photo Picker (no permission); else granular `READ_MEDIA_*` + partial (14).
17. *Location?* Fg/bg split, approximate vs precise, "only this time"; least privilege; background separate/after (Ch 67).

Security/UX

18. *Least privilege?* Minimal permission (coarse, Photo Picker) — privacy + Play.
19. *Play justification?* Declared/justified sensitive use or rejection.
20. *Respect "no"?* Degrade, don't coerce/nag.

Testing/Production

21. *Test the state machine?* Unit test the 4 states.
22. *Test real dialogs?* UiAutomator / `GrantPermissionRule` .
23. *Low grant rate cause?* Out-of-context / weak rationale.
24. *Feature "broken"?* Permanent denial not routed to settings.
25. *Crashes?* Not checking at use.
26. *Key metric?* Grant funnel + permanent-denial rate.

Design-change challenges

1. **Camera scan feature with respectful permission UX.** In-context + rationale + settings on permanent denial + degrade.
2. **Enable notifications on Android 13+.** `POST_NOTIFICATIONS` after onboarding; inbox resilient if denied.
3. **Add a review photo without full gallery access.** Photo Picker (no permission).
4. **App crashes when a user revokes location in settings.** Check at use; degrade.
5. **Feature permanently dead after "don't ask again".** Detect permanent denial → settings deep-link.

Mock walkthrough (abridged)

Clarify (which permissions, can we avoid them, when triggered?) → prefer no-permission APIs (Photo Picker/approximate) → in-context request + rationale → 4-state machine (`shouldShowRationale`) → permanent denial → settings deep-link (Ch 52) → check at use + re-check on resume → degrade gracefully → least privilege + Play justification (Ch 28) → testing (UiAutomator) → grant-funnel observability.

51.17 Practical Exercises

1. **Beginner.** Request `CAMERA` in context via the Activity Result API when the user taps "Scan"; open the scanner on grant. *Eval:* in-context; granted → scanner.
2. **Intermediate.** Implement the 4-state machine: rationale on denied-can-ask, settings deep-link on permanent denial, degrade on denial. *Eval:* each state routed; feature recoverable; no crash.
3. **Advanced.** Replace a `READ_MEDIA_IMAGES` flow with the Photo Picker (no permission) and request `POST_NOTIFICATIONS` after onboarding. *Eval:* no media permission; notifications requested in context; inbox works if denied.
4. **Production debugging.** Fix a crash when location is revoked (check at use) and a "dead feature" after permanent denial (settings deep-link). *Eval:* no crash; recoverable.
5. **System design challenge.** Build a reusable `PermissionController` / `FeatureGate` and a grant-funnel dashboard; A/B two rationales. *Eval:* reused across features; measurable grant rate.

(Solutions withheld unless requested.)

51.18 Chapter Summary & Reference Sheet

Key decisions: request dangerous permissions **in context** (on the feature trigger, never at launch) with a clear **rationale**, using the **Activity Result API** (`registerForActivityResult` / `rememberLauncherForActivityResult` , not the deprecated callback); model the **full 4-state machine — not-asked / granted / denied-can-ask (rationale + re-request) / permanently-denied (settings deep-link, no silent re-request)** — distinguishing the last two via `shouldShowRequestPermissionRationale` (persist an "alreadyAsked" flag to tell not-asked from permanent); **check the permission at the point of use** (it can be revoked or auto-reset) and re-check on resume; **degrade gracefully** (disable only that feature, never crash/block/nag); and **prefer least-privilege, no-permission modern APIs** — the **Photo Picker** for media, **approximate** location, **"only this time"**, and request **background location only separately** (Ch 67). Notifications are a runtime permission on Android 13+, but the inbox (Ch 50) must still work if denied.

Concepts: dangerous permissions, Activity Result API, 4-state machine, `shouldShowRequestPermissionRationale`, check-at-use, auto-reset, graceful degradation, settings deep-link, notifications (13+), location (fg/bg/approximate/one-time), media (granular/Photo Picker/partial), least privilege.

Common mistakes: asking at launch (low grants); treating denial as one state; re-requesting when permanently denied (silent no-op); caching "granted" (crash on revoke); blocking the whole app / nagging on denial; over-broad requests (full media, precise/background location) hurting grants and Play approval; using deprecated APIs.

Checklist:

- [ ] request in context
- [ ] Activity Result API
- [ ] 4-state machine via `shouldShowRationale` + persisted "alreadyAsked"
- [ ] rationale UI
- [ ] settings deep-link on permanent denial (Ch 52)
- [ ] check at use + re-check on resume
- [ ] degrade just the feature
- [ ] Photo Picker / approximate location (least privilege)
- [ ] notifications in context (inbox resilient)
- [ ] grant-funnel telemetry.

Key APIs: `ContextCompat.checkSelfPermission`, `ActivityResultContracts.RequestPermission` / `RequestMultiplePermissions` / `PickVisualMedia`, `registerForActivityResult` / `rememberLauncherForActivityResult`, `Activity.shouldShowRequestPermissionRationale`, `Settings.ACTION_APPLICATION_DETAILS_SETTINGS` (Ch 52), `POST_NOTIFICATIONS`, `ACCESS_FINE/COARSE/BACKGROUND_LOCATION`, `READ_MEDIA_IMAGES/VIDEO/AUDIO` / `READ_MEDIA_VISUAL_USER_SELECTED`.

References:

- Request runtime permissions: https://developer.android.com/training/permissions/requesting
- Permissions best practices: https://developer.android.com/training/permissions/usage-notes
- Notification permission: https://developer.android.com/develop/ui/views/notifications/notification-permission
- Photo Picker: https://developer.android.com/training/data-storage/shared/photopicker
- Location permissions: https://developer.android.com/develop/sensors-and-location/location/permissions
- (composes) Ch 44 Lifecycle, Ch 49/50 Notifications, Ch 52 Deep Links (settings), Ch 67 Location, Ch 28 Privacy

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/media/failure diagrams; state model + controller + Compose in-context flow + notifications + Photo Picker + location code; concurrency (check-at-use, resume re-check, lifecycle-safe launcher); failure (grant-rate/permanent-denial/revoke triage); 5 levels; in-context, Photo-Picker, approximate-location trade-offs; security (least privilege, Play justification, auto- reset); perf (avoid permissions); unit + UiAutomator tests; observability (grant funnel); 26 Q&A + 5 challenges. ✅
- **Expand later:** location (Ch 67), notifications (Ch 49/50), deep links to settings (Ch 52), privacy (Ch 28).
- **Interview coverage:** exceeds ≥25 bar.